

D767

Elementary Literacy and Mathematics Strategies and Assistive Technologies

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A third grader with a math disability reads this problem: “Maya had some stickers. She gave away 6 and has 9 left. How many did she start with?” He writes $9 - 6 = 3$. He attends a school with a strong chess club, and his IEP meeting is next month. What should the teacher teach?

- A. Teach a list of key words, such as “left” means subtract.
- B. Wait to address it at next month’s IEP meeting.
- C. Assign ten more subtraction problems for practice.
- D. Identify the problem as a change type and place the numbers in a start-change-end diagram before choosing the operation.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which word problem is a compare problem?

- A. Ana had 12 shells and found 5 more. How many does she have now?
- B. Ana had 12 shells and gave 5 away. How many are left?
- C. Ana has 12 shells. Ben has 5 fewer shells than Ana. How many shells does Ben have?
- D. Ana has 12 shells and 5 rocks. How many objects does she have?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A fourth grader’s worksheet shows $52 - 38 = 26$, $61 - 27 = 46$, and $43 - 19 = 36$. She finished it quickly, sitting near the class aquarium, and wears glasses. What is the most likely error pattern?

- A. She is adding instead of subtracting.
- B. She cannot see the numbers clearly.
- C. She subtracts the smaller digit from the larger in each column instead of regrouping.
- D. She makes random careless errors from rushing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A student always subtracts the smaller digit from the larger digit in each column. What answer will he most likely give for $71 - 49$?

- A.** 120
- B.** 38
- C.** 32
- D.** 22

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A second grader reads 20 words correct per minute at baseline. The team uses an ambitious growth rate of 1.5 words per week. The school year has 30 weeks left, and the student also receives speech services twice a week. What is the year-end goal?

- A.** 65 words correct per minute
- B.** 23 words correct per minute
- C.** 45 words correct per minute
- D.** 600 words correct per minute

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A student's baseline is 30 words correct per minute. The IEP goal is 72 words correct per minute in 28 weeks. What weekly growth rate does the goal assume?

- A.** 2.6 words per week
- B.** 1.5 words per week
- C.** 42 words per week
- D.** 3.6 words per week

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

After a read-aloud of a folk tale, a third grader with a language disability retells two favorite details out of order and leaves out the problem. The class is on a field trip next week, and the book had bright illustrations. What should the teacher teach?

- A. Give a vocabulary quiz on the tale's hard words.
- B. Have him describe the illustrations in detail.
- C. Have him reread the tale three times.
- D. A story map with self-questions for characters, setting, problem, events, and solution.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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8.

A fifth grader with a learning disability writes two-sentence opinion pieces and says, "I'm done." Her teacher has a colorful planning sheet and the school has a new library. Which approach reflects SRSD?

- A. Require a minimum of 300 words every time.
- B. Let her dictate every essay to an adult.
- C. Model a planning strategy with self-talk, then have her set a goal for parts included, monitor, and graph her progress.
- D. Hand her the colorful planning sheet and ask for longer essays.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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9.

A fourth grader tells detailed stories aloud but writes only a few misspelled words in ten minutes. He is the tallest in his class and plays soccer on weekends. Which assistive technology best matches his barrier?

- A. A graphic organizer for ideas.
- B. Speech-to-text, so he can dictate and then edit his drafts.
- C. A text-to-speech reader.
- D. A spell checker only.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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10.

A fifth grader with a math disability has an IEP goal to solve two-step word problems. Her computation is slow, and she recently joined the school band. Last year's teacher never allowed calculators. During word-problem practice, what is the best decision?

- A. Allow a calculator so her effort goes to setting up and solving the problem.
- B. Remove word problems until her computation improves.
- C. Ban the calculator, as last year's teacher did.
- D. Never allow calculators until facts are mastered.

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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11. [TRANSFER — reversed stem / different variable]

During a lesson whose goal is fluent recall of multiplication facts through 10×10 , a student with an IEP asks to use a calculator. What is the best answer?

- A. No, and remove calculators from his word-problem work too.
- B. Yes, but only for facts above 5×5 .
- C. Yes, because calculators are listed accommodations for all math.
- D. Not during this lesson, because the calculator would do the skill being taught.

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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12.

A kindergarten teacher uses Elkonin boxes with the word "ship." The class pet is a hamster and the room has a new carpet. How many boxes should the word have?

- A. 3
- B. 2
- C. 4
- D. 1

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A third grader with a math disability reads this problem: “Maya had some stickers. She gave away 6 and has 9 left. How many did she start with?” He writes $9 - 6 = 3$. ~~He attends a school with a strong chess club, and his IEP meeting is next month.~~ What should the teacher teach?

Target word: fix for word-problem errors **Decides:** He chose subtraction from the word “left”; it is a change problem with an unknown start.

A	Inverted operation	That key-word rule is what produced the wrong answer here.
B	Camouflage	The meeting date is irrelevant to what to teach now.
C	Dropped element	Drops the problem-type analysis; more subtraction reinforces the error.
D	Correct	The schema shows the start is unknown, so $6 + 9 = 15$.

2. Answer C Concept 1 • transfer

Which word problem is a compare problem?

Target word: recognize the compare schema **Decides:** Compare = two separate sets and the difference between them.

A	Confusion set	One set changes over time: a change problem.
B	Confusion set	A change problem with a decrease, not a comparison.
C	Correct	Two sets are compared by a difference; that is the compare schema.
D	Adjacent category	Parts combine into a whole: a group/total problem.

3. Answer C Concept 2

A fourth grader’s worksheet shows $52 - 38 = 26$, $61 - 27 = 46$, and $43 - 19 = 36$. She finished it quickly, ~~sitting near the class aquarium, and wears glasses.~~ What is the most likely error pattern?

Target word: the consistent error **Decides:** In each ones column she subtracts the smaller digit from the larger.

A	Inverted operation	$52 + 38 = 90$, not 26; the answers are not sums.
B	Camouflage	The glasses detail does not explain a consistent procedure.
C	Correct	$8 - 2 = 6$, $7 - 1 = 6$, $9 - 3 = 6$ in the ones; the pattern skips regrouping.
D	Adjacent category	The errors follow one rule, so they are not random.

4. Answer B Concept 2 • transfer

A student always subtracts the smaller digit from the larger digit in each column. What answer will he most likely give for $71 - 49$?

Target word: predict the answer from the bug **Decides:** Ones: $9 - 1 = 8$; tens: $7 - 4 = 3$.

A	Inverted operation	That is the sum, not the buggy difference.
B	Correct	Applying the smaller-from-larger rule gives 3 tens and 8 ones.
C	Confusion set	Swaps the tens and ones digits of the buggy result.
D	Dropped element	That is the correct difference; the question asks what the error pattern produces.

5. Answer A Concept 3

A second grader reads 20 words correct per minute at baseline. The team uses an ambitious growth rate of 1.5 words per week. The school year has 30 weeks left, ~~and the student also receives speech services twice a week.~~ What is the year-end goal?

Target word: end-of-year goal **Decides:** $20 + (1.5 \times 30) = 65$.

A	Correct	Baseline 20 + 45 words of growth = 65.
B	Camouflage	Adds the twice-weekly speech sessions and growth rate as if they matter; they do not.
C	Dropped element	Growth only; the baseline was not added.
D	Inverted operation	Multiplies baseline by weeks instead of adding growth.

6. Answer B Concept 3 • transfer

A student's baseline is 30 words correct per minute. The IEP goal is 72 words correct per minute in 28 weeks. What weekly growth rate does the goal assume?

Target word: solve for growth rate **Decides:** $(72 - 30) / 28 = 1.5$.

A	Dropped element	$72 / 28$ ignores the baseline.
B	Correct	42 words of growth over 28 weeks = 1.5 per week.
C	Wrong stem number	Total growth, not per week.
D	Inverted operation	$(72 + 30) / 28$ adds baseline instead of subtracting it.

7. Answer D Concept 4

After a read-aloud of a folk tale, a third grader with a language disability retells two favorite details out of order and leaves out the problem. ~~The class is on a field trip next week, and the book had bright illustrations.~~ What should the teacher teach?

Target word: strategy for narrative comprehension **Decides:** He lacks a structure for story parts.

A	Adjacent category	Vocabulary helps, but the gap shown is story structure.
B	Camouflage	Focuses on pictures, not story structure.
C	Dropped element	Rereading without a structure does not teach him what to look for.
D	Correct	Story grammar gives a frame to organize and recall the story.

8. Answer C Concept 5

A fifth grader with a learning disability writes two-sentence opinion pieces and says, "I'm done." ~~Her teacher has a colorful planning sheet and the school has a new library.~~ Which approach reflects SRSD?

Target word: what SRSD adds **Decides:** SRSD = explicit strategy plus self-regulation, taught in stages.

A	Absolute language	A length rule without strategy does not teach planning.
B	Adjacent category	Dictation addresses transcription, not planning and persistence.
C	Correct	Explicit modeling plus goal setting, self-talk, and self-monitoring are the core of SRSD.
D	Camouflage	An organizer alone lacks the modeling and self-regulation.

9. Answer B Concept 6

A fourth grader tells detailed stories aloud but writes only a few misspelled words in ten minutes. ~~He is the tallest in his class and plays soccer on weekends.~~ Which assistive technology best matches his barrier?

Target word: match AT to a transcription barrier **Decides:** Ideas are strong; getting words onto the page is the barrier.

A	Adjacent category	His ideas are already strong; the barrier is transcription.
B	Correct	Removes the transcription barrier so his ideas reach the page.
C	Confusion set	Text-to-speech reads text aloud; it supports reading, not writing output.
D	Dropped element	He produces too few words for spell checking to help much.

10. Answer A Concept 7

A fifth grader with a math disability has an IEP goal to solve two-step word problems. Her computation is slow, and ~~she recently joined the school band. Last year's teacher never allowed calculators.~~ During word-problem practice, what is the best decision?

Target word: calculator use for a problem-solving goal **Decides:** The target skill is reasoning, not computation.

A	Correct	The calculator removes a computation barrier without changing the skill measured.
B	Inverted operation	Withholds the goal skill instead of removing the barrier.
C	Camouflage	Past practice does not decide fit with the current goal.
D	Absolute language	A blanket rule ignores which skill is targeted.

11. Answer D Concept 7 · transfer

During a lesson whose goal is fluent recall of multiplication facts through 10×10 , a student with an IEP asks to use a calculator. What is the best answer?

Target word: when a calculator is not an accommodation **Decides:** The target skill is computation itself.

A	Inverted operation	Over-applies the rule to a problem-solving task where the calculator fits.
B	Wrong stem number	Splits the fact set arbitrarily; all facts are the target.
C	Absolute language	An accommodation must not replace the skill being taught.
D	Correct	A support that performs the target skill changes what is learned.

12. Answer A Concept 8

A kindergarten teacher uses Elkonin boxes with the word "ship." ~~The class pet is a hamster and the room has a new carpet.~~ How many boxes should the word have?

Target word: boxes = phonemes **Decides:** ship = /sh/ /i/ /p/: three sounds.

A	Correct	The digraph sh is one sound, so there are three phonemes.
B	Dropped element	Drops one of the sounds.
C	Confusion set	Counts letters, not sounds.
D	Adjacent category	Counts syllables; Elkonin boxes mark phonemes.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Schema-based word-problem instruction	Students learn to spot the problem type (group/total, change, compare) and map the numbers onto a diagram for that type before choosing an operation.	Teaching key words such as “left” means subtract or “in all” means add.	A student picks operations from single words in the problem.
2	Error analysis in computation	Look for a consistent wrong procedure across items; a pattern points to a specific misconception to reteach.	Counting total errors and assigning more of the same practice.	A worksheet where the wrong answers follow one rule.
3	Setting a goal from baseline and growth rate	Goal = baseline + (weekly growth rate × number of weeks); the goal line connects baseline to that end point.	Multiplying the baseline by the weeks or forgetting to add the baseline.	A baseline score, a weekly growth rate, and a number of weeks.
4	Story maps and self-questioning	Teaching students to ask and answer story-grammar questions (who, where, problem, events, solution) organizes recall and comprehension of narrative text.	Having students reread the story until they remember it.	A student retells random details with no sequence or problem.
5	SRSD for writing	Self-Regulated Strategy Development teaches a writing strategy explicitly plus self-regulation (goal setting, self-talk, self-monitoring) through stages from modeling to independent use.	Giving a graphic organizer alone and expecting better essays.	Students write short, unplanned pieces and quit early.
6	AT for written expression	Speech-to-text and word prediction reduce transcription barriers (spelling, handwriting) so a student can show ideas; the tool matches the barrier.	Choosing a spell checker for a student whose barrier is getting any words down at all.	A student says strong ideas aloud but produces few written words.
7	Calculator: accommodation or not	A calculator is a fair accommodation when the skill being taught or measured is problem solving, but not when the skill is computation itself.	Allowing or banning calculators everywhere as a blanket rule.	The IEP goal or lesson target names computation vs. reasoning.
8	Elkonin boxes for phoneme segmentation	The student pushes one marker into one box for each sound (not each letter), making phonemes visible.	Drawing one box per letter.	A word with a digraph or silent letter.

Course scope (public description): Elementary Literacy and Mathematics Strategies and Assistive Technologies prepares special education teachers to use progress monitoring and evidence-based instructional practices appropriate for elementary students with exceptionalities. The course includes cognitive and metacognitive strategies that elementary students can use to acquire new content knowledge and generalize skills across learning environments. It also provides opportunities for learners to incorporate intensive instructional strategies and practice making accommodations that include assistive technologies to elementary math and English language arts lesson plans based on learner characteristics, performance data, and individualized education

program (IEP) goals.